MATEMATICI 2

Seminar 5

Ex. Fie $f : \mathbb{R}^3 \to \mathbb{R}^3$

$f(x_1, x_2, x_3) = (-2x_1 + x_2 + x_3 \ , \ x_1 - 2x_2 + x_3 \ , \ x_1 + x_2 - 2x_3)$

Să se det. $M_f^{B \text{ canonică}}$

bază canonică $B = \{ e_1 = (1,0,0) \ , \ e_2 = (0,1,0) \ , \ e_3 = (0,0,1) \}$

Fie $v \in \mathbb{R}^3$; $v = (x_1, x_2, x_3)$

$$f(v) = f(x_1, x_2, x_3) = (-2x_1 + x_2 + x_3,\ x_1 - 2x_2 + x_3,\ x_1 + x_2 - 2x_3)$$

$$f(x_1, x_2, x_3) = (-2x_1 + x_2 + x_3) \cdot (1,0,0) + (x_1 - 2x_2 + x_3) \cdot (0,1,0) + (x_1 + x_2 - 2x_3) \cdot (0,0,1)$$

$\det. M_f^B$

$$f(e_1) = f(1,0,0) = (-2, 1, 1) = -2e_1 + e_2 + e_3$$

$$f(e_2) = f(0,1,0) = (1, -2, 1) = e_1 - 2e_2 + e_3$$

$$f(e_3) = f(0,0,1) = (1, 1, -2) = e_1 + e_2 - 2e_3$$

$$M_f^B = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} = A$$! coord. se pun pe coloane!

det. val. proop.

$$\det(A-\lambda I_3) = \det\left(\begin{pmatrix}-2 & 1 & 1\\ 1 & -2 & 1\\ 1 & 1 & -2\end{pmatrix} - \begin{pmatrix}\lambda & 0 & 0\\ 0 & \lambda & 0\\ 0 & 0 & \lambda\end{pmatrix}\right) =$$

$$= \begin{vmatrix}-2-\lambda & 1 & 1\\ 1 & -2-\lambda & 1\\ 1 & 1 & -2-\lambda\end{vmatrix} = (-2-\lambda)^3 + 1 + 1 + 2+\lambda + 2+\lambda + 2+\lambda$$

$$= -(2+\lambda)^3 + 2 + 6 + 3\lambda = -(8 + 12\lambda + 6\lambda^2 + \lambda^3) + 8 + 3\lambda$$

$$= -8 - 12\lambda - 6\lambda^2 - \lambda^3 + 8 + 3\lambda$$

$$= -\lambda^3 - 6\lambda^2 - 9\lambda = -\lambda(\lambda^2 + 6\lambda + 9) =$$

$$= -\lambda(\lambda+3)^2 = 0$$

$$\begin{cases} \lambda_1 = 0 & n_1 = 1 \\ \lambda_2 = -3 & n_2 = 2 \end{cases}$$

$$V_{\lambda_1} = \{ a(1,0,0) + b(0,1,0) + c(0,0,1) \mid a,b,c \in \mathbb{R}, (M_f^B - \lambda I_3)\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$$

$$V_{\lambda_1} = \{ (a,b,c) \mid a,b,c \in \mathbb{R}, (M_f^B - \lambda_1 I_3)\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$$

$$M_f^B - \lambda_1 I_3 = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} - \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_1 \end{pmatrix} = \begin{pmatrix} -2-\lambda_1 & 1 & 1 \\ 1 & -2-\lambda_1 & 1 \\ 1 & 1 & -2-\lambda_1 \end{pmatrix}$$

$$= M_f^B = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -2a+b+c \\ a-2b+c \\ a+b-2c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$a+b = 2c \Rightarrow c = \frac{a+b}{2} \Rightarrow \boxed{c = a = b}$$

$$-2a+b+\frac{a+b}{2} = 0 \quad / \cdot 2$$

$$-4a+2b+a+b = 0$$

$$-3a+3b = 0$$

$$-a+b = 0 \Rightarrow \boxed{a = b}$$

$V_{\lambda_1} = \{(a, a, a) \mid a \in \mathbb{R}\}$ $\{(1,1,1)\}$ bază pt. V_{λ_1}

$$M_f^B - \lambda_2 I_3 = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} - \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a+b+c \\ a+b+c \\ a+b+c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$a+b+c = 0 \Rightarrow c = -a-b$

$$V_{\lambda_2} = \{(a, b, -(a+b)) \mid a, b \in \mathbb{R}\}$$

$$\begin{matrix} b = \alpha \\ c = \beta \end{matrix} \Rightarrow a = -b - c = -(\alpha + \beta)$$

$$\Rightarrow (x, y, z) = \alpha \underbrace{(-1, 1, 0)}_{V_1} + \beta \underbrace{(-1, 0, 1)}_{V_2} \Rightarrow$$

$\Rightarrow V_{h_1} = \{ \alpha v_1 + \beta v_2 \mid \alpha, \beta \in \mathbb{R} \} \Rightarrow$

$\Rightarrow \{v_1, v_2\}$ sist. generatori

Din $\alpha v_1 + \beta v_2 = (0,0,0) \Rightarrow \begin{cases} \alpha = 0 \\ \beta = 0 \end{cases} \Rightarrow \{v_1, v_2\}$ liniar indep.

Ex.

Fie $f : V \to V$, V sp. vect. real finit, f liniară

$f \circ f = f$

Să se det. val. proprii și subsp de vect. prop

Rez.

Obs. f sn. proiector

Fie $v \in V$, $v \neq 0$

„$\Rightarrow$" $f(f(v)) = f(v) = \lambda v =$

$= f(\lambda v) = \lambda f(v) = \lambda^2 v$

$\lambda^2 v - \lambda v = \lambda(\lambda - 1)v = 0 \Rightarrow$ $\lambda_1 = 0$, $\lambda_2 = 1$

Cum $v \neq 0$

$V_{\lambda_1} = \{ v \in V \mid f(v) = \lambda_1 v \} = \ker f$

$f(v) = 0 \cdot v = 0$

$f(f(v)) = f(0 \cdot v) = 0 \cdot f(v) = 0$

$f(0) = 0$

$V_{\lambda_2} = \{ v \in V \mid f(v) = \lambda_2 v \}$

$f(v) = 1 \cdot v = v$

$f(v) = v \Rightarrow v \in Imf \Rightarrow V_{\lambda_2} \subseteq Imf$ (1)

Fie $f(v) \in Imf$

$f(f(v)) = 1 \cdot f(v) \in Imf \Rightarrow$

$\Rightarrow f(v) \in V_{\lambda_2} = \{\omega \in V \mid f(\omega) = 1 \cdot \omega\} \Rightarrow$

$\Rightarrow Imf \subseteq V_{\lambda_2}$ (2)

Din (1) și (2) $\Rightarrow Imf = V_{\lambda_2}$